

Question 1

Figure below shows a series–shunt feedback amplifier known as a “feedback triple.” All three MOSFETs are biased to operate at $g_m = 4 \text{ mA/V}$. You may neglect their r_o 's.

- Select a value for R_F that results in a closed-loop gain that is ideally 10 V/V .
- Determine the loop gain $A\beta$ and hence the value of A_f . By what percentage does A_f differ from the ideal value you designed for? How can you adjust the circuit to make A_f equal to 10?

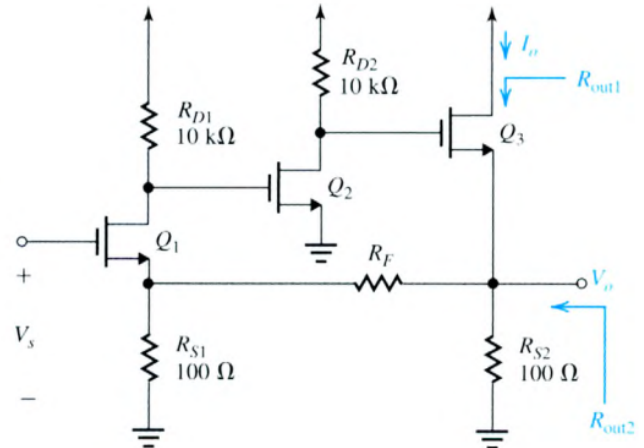
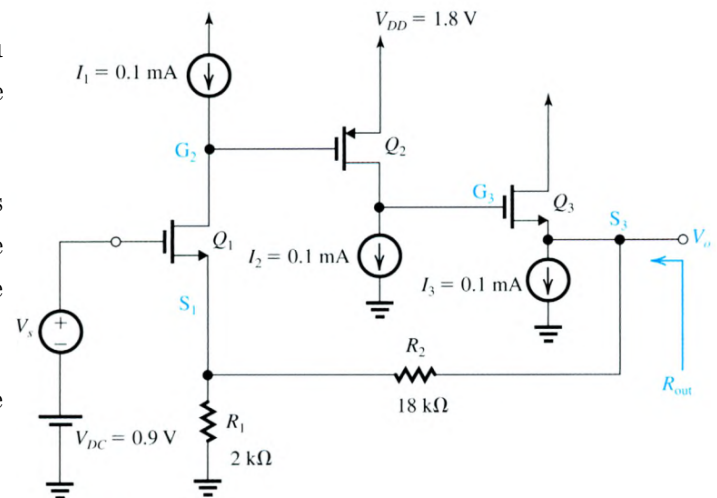
**Question 2**

Figure below shows a series–shunt amplifier in which the three MOSFETs are sized to operate at $|V_{OV}| = 0.2 \text{ V}$. Let $|V_t| = 0.5 \text{ V}$ and $|V_A| = 10 \text{ V}$. The current sources utilize single transistors and thus have output resistances equal to r_o .

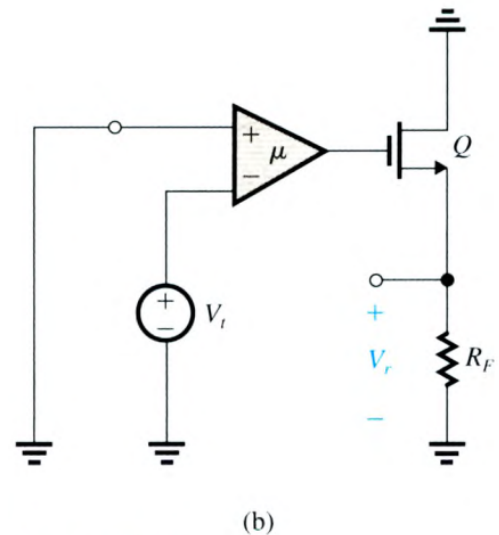
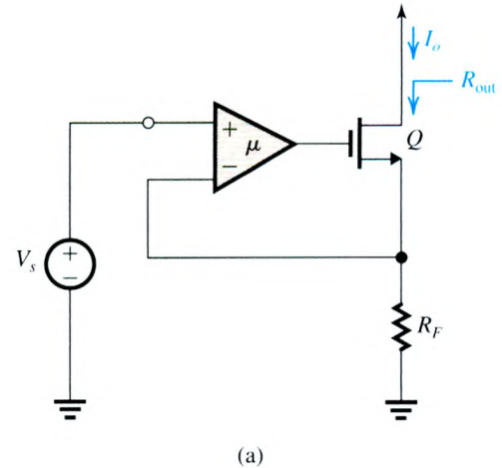
- Show that the feedback is negative.
- Assuming the loop gain to be large, what do you expect the closed-loop voltage gain V_o/V_s to be approximately?
- If V_s has a zero dc component, find the dc voltages at nodes S_1 , G_2 , S_3 , and G_3 . Verify that each of the current sources has the minimum required dc voltage across it for proper operation.
- Find the A circuit. Calculate the gain of each of the three stages and the overall voltage gain, A.
- Find β .
- Find $A_f = V_o/V_s$. By what percentage does this value differ from the approximate value obtained in (b)?
- Find the output resistance R_{out} .



Question 3

Figure (a) below shows a feedback transconductance amplifier using an op amp together with an NMOS transistor. The feedback network consists of a resistor R_F that senses the output current I_o (recall that the drain and source currents of a MOSFET are equal) and provides a feedback voltage that is subtracted from V_s by means of the differencing action of the op-amp input.

- Setting $A=\infty$, find the ideal closed-loop gain I_o/V_s .
- Find the value of R_F that results in an ideal closed-loop gain of 10 mA/V.
- Find an expression for the loop gain $A\beta$ by setting $V_s=0$, breaking the loop, and applying an input test voltage V_t , as shown in figure (b). Assume the input resistance of the op amp to be very high.
- For $\mu=1000$, and assuming the MOSFET has $g_m=2\text{mA/V}$ and $r_o=20\text{ k}\Omega$, find the values of $A\beta$, A , and A_f .



Question 4

For the feedback transconductance amplifier of figure below, derive expressions for A , β , $A\beta$, A_f , R_o , and R_{of} . Evaluate A_f and R_{of} for the case of $g_{m1}=g_{m2}=4\text{ mA/V}$, $R_D=25\text{ k}\Omega$, $r_{o2}=20\text{ k}\Omega$, $R_F=100\text{ }\Omega$, and $R_L=1\text{ k}\Omega$. For simplicity, neglect r_{o1} and take r_{o2} into account only when calculating output resistances. Also, evaluate the output resistance R_{out} seen between the two terminals of R_L .

